

Лабораторная работа №7

Дисциплина: Администрирование сетевых подсистем

Бахи сиди али темассини

2026-02-16

1. Цель работы
2. Выполнение лабораторной работы
3. Выводы

Раздел 1

1. Цель работы

1.1 Цель работы

- Получение навыков настройки межсетевого экрана в Linux

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- Получение навыков настройки межсетевого экрана в Linux
- Реализация переадресации портов

1.1 Цель работы

- Получение навыков настройки межсетевого экрана в Linux
- Реализация переадресации портов
- Настройка Masquerading

Раздел 2

2. Выполнение лабораторной работы

2.1 Создание пользовательской службы firewalld

- Выполнен переход в режим суперпользователя

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/ssh-custom.xml
[root@server.bahis.net ~]# cd /etc/firewalld/services/
[root@server.bahis.net services]# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing commands on remote machines.
It provides secure encrypted communications. If you plan on accessing your machine remotely via SSH over a f
irewalled interface, enable this option. You need the openssh-server package installed for this option to be
useful.</description>
  <port protocol="tcp" port="22"/>
</service>
```

Рисунок 1: Копирование файла службы ssh.xml в ssh-custom.xml

2.1 Создание пользовательской службы firewalld

- Выполнен переход в режим суперпользователя
- Скопирован файл `ssh.xml` в `ssh-custom.xml`

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/ssh-custom.xml
[root@server.bahis.net ~]# cd /etc/firewalld/services/
[root@server.bahis.net services]# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing commands on remote machines.
It provides secure encrypted communications. If you plan on accessing your machine remotely via SSH over a f
irewalled interface, enable this option. You need the openssh-server package installed for this option to be
useful.</description>
  <port protocol="tcp" port="22"/>
</service>
```

Рисунок 1: Копирование файла службы `ssh.xml` в `ssh-custom.xml`

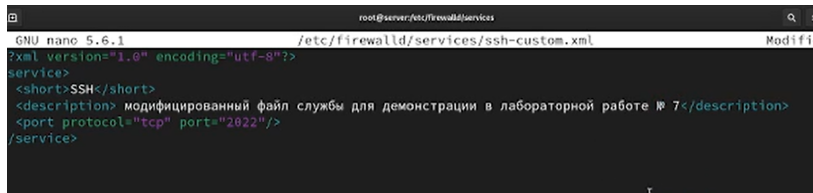
2.1 Создание пользовательской службы firewalld

- Выполнен переход в режим суперпользователя
- Скопирован файл `ssh.xml` в `ssh-custom.xml`
- Файл размещён в `/etc/firewalld/services/`

```
[bahis@server.bahis.net ~]$ sudo -i
[sudo] password for bahis:
[root@server.bahis.net ~]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/ssh-custom.xml
[root@server.bahis.net ~]# cd /etc/firewalld/services/
[root@server.bahis.net services]# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing commands on remote machines.
It provides secure encrypted communications. If you plan on accessing your machine remotely via SSH over a f
irewalled interface, enable this option. You need the openssh-server package installed for this option to be
useful.</description>
  <port protocol="tcp" port="22"/>
</service>
```

Рисунок 1: Копирование файла службы `ssh.xml` в `ssh-custom.xml`

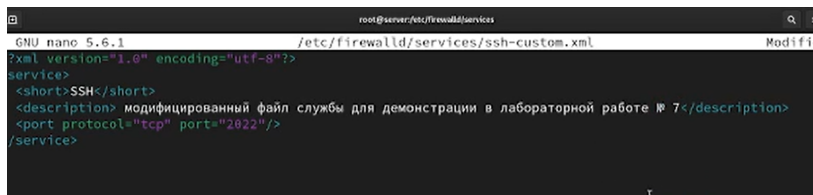
- В файле изменён порт с 22 на 2022



```
root@server:/etc/firewalld/services
GNU nano 5.6.1 /etc/firewalld/services/ssh-custom.xml
?xml version="1.0" encoding="utf-8"?>
service
  <short>SSH</short>
  <description> модифицированный файл службы для демонстрации в лабораторной работе № 7</description>
  <port protocol="tcp" port="2022"/>
/service
```

Рисунок 2: Редактирование службы: изменение порта на 2022 и обновление описания

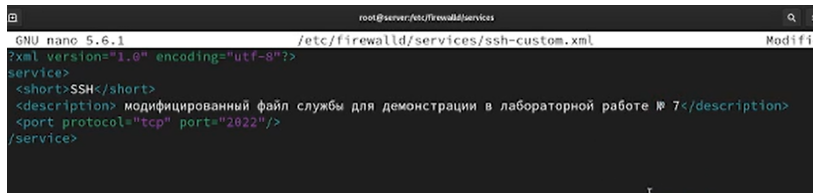
- В файле изменён порт с 22 на 2022
- Обновлено описание службы



```
root@server:/etc/firewalld/services
GNU nano 5.6.1 /etc/firewalld/services/ssh-custom.xml Modified
?xml version="1.0" encoding="utf-8"?>
service>
  <short>SSH</short>
  <description> модифицированный файл службы для демонстрации в лабораторной работе № 7</description>
  <port protocol="tcp" port="2022"/>
/service>
```

Рисунок 2: Редактирование службы: изменение порта на 2022 и обновление описания

- В файле изменён порт с 22 на 2022
- Обновлено описание службы
- Проверено содержимое файла



```
root@server:/etc/firewalld/services
GNU nano 5.6.1 /etc/firewalld/services/ssh-custom.xml Modified
?xml version="1.0" encoding="utf-8"?>
service>
  <short>SSH</short>
  <description> модифицированный файл службы для демонстрации в лабораторной работе № 7</description>
  <port protocol="tcp" port="2022"/>
/service>
```

Рисунок 2: Редактирование службы: изменение порта на 2022 и обновление описания

● Получен список служб --get-services

```
root@server:/etc/firewall/services
[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-t
estnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collect
d condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-
tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman forema
n-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gan
glia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ips
ec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api
kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager
-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kub
elet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesi
eve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula net
bios-nfs netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storagec
onsole ovirt-vmconsole plex pcmd pmpoxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometh
eus-node-exporter proxy-dhcp ps2link ps3netstrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-se
ntinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sip6 smtp
smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh stea
m-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp
tile38 tinc tor-socks transmission-client upnp-client vds vnc vnc-server warpinator wbem-http wdem-https wireg
uard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmpc xmpp-bosh xmpp-cl
ient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server.bahis.net services]# firewall-cmd --reload
success
[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-t
estnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collect
d condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-
tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman forema
n-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gan
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ec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api
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m-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp
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ient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
```

- Получен список служб --get-services
- Выполнена перезагрузка firewall-cmd --reload

```

root@server:/etc/firewall/services
[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
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[root@server.bahis.net services]# firewall-cmd --reload
success
[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
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ient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier

```

- Получен список служб --get-services
- Выполнена перезагрузка firewall-cmd --reload
- Служба ssh-custом добавлена в активные

```

root@server:/etc/firewall/services
[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-t
estnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collect
d condor-collector cratedb ctddb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-
tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman forema
n-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gan
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elet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesi
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smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh stea
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[root@server.bahis.net services]# firewall-cmd --reload
success
[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
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tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman forema
n-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gan
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ec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api
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eve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula net
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onsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometh
eus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-se
ntinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp
smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh stea
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ient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier

```


- Получен список служб --get-services
- Выполнена перезагрузка firewall-cmd --reload
- Служба ssh-custом добавлена в активные
- Добавлена в постоянную конфигурацию

```

root@server:/etc/firewall/services
[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-t
estnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collect
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n-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gan
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ec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api
kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager
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bios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storagec
onsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometh
eus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-se
ntinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp
smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh stea
m-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp
tile38 tinc tor-socks transmission-client upnp-client vdsms vnc-server warpinator wbem-http wdem-http wireg
uard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmpc xmpp-bosh xmpp-cl
ient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server.bahis.net services]# firewall-cmd --reload
success
[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-t
estnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collect
d condor-collector cratedb ctddb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-
tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman forema
n-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gan
glia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ips
ec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api
kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager
-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kub
elet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesi
eve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula net
bios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storagec
onsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometh
eus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-se
ntinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp
smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh stea
m-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle tftp
tile38 tinc tor-socks transmission-client upnp-client vdsms vnc-server warpinator wbem-http wdem-http wireg
uard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmpc xmpp-bosh xmpp-cl
ient xmpp-local xmpp-server zabbix-agent zabbix-server zerotier

```

```

[root@server.bahis.net services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisapp
bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-t
estnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit collect
d condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-
tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger foreman forema
n-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client gan
glia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ips
irc ircs iscsi-target isns jenkins kadmin kdeconnect keyberos kibana klogin kpasswd kprop kshell kube-api
kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager
-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kub
elet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesi
eve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula net
bios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storagec
onsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometh
eus-node-exporter proxy-dhcp ps2link ps3netdrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-se
ntinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp
smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-
custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet te
ntacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsms vnc-server warpinator wbem-http wbe
https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-b
osh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server.bahis.net services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https ssh
[root@server.bahis.net services]# firewall-cmd --add-service=ssh-custom
success
[root@server.bahis.net services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns http https ssh ssh-custom
[root@server.bahis.net services]# firewall-cmd --add-service=ssh-custom --permanent
success
[root@server.bahis.net services]# firewall-cmd --reload
success
[root@server.bahis.net services]# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22

```

Рисунок 4: Добавление и активация пользовательской службы ssh-custom в firewalld

2.2 Перенаправление портов

- Настроено правило перенаправления:

```
[root@server.bahis.net services]# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22  
success
```

Рисунок 5: Настройка перенаправления порта 2022 на 22 в firewalld

2.2 Перенаправление портов

- Настроено правило перенаправления:
- 2022 → 22 (TCP)

```
[root@server.bahis.net services]# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22  
success
```

Рисунок 5: Настройка перенаправления порта 2022 на 22 в firewalld

2.2 Перенаправление портов

- Настроено правило перенаправления:
- 2022 → 22 (TCP)
- Правило добавлено в текущую конфигурацию

```
[root@server.bahis.net services]# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22  
success
```

Рисунок 5: Настройка перенаправления порта 2022 на 22 в firewalld

- С клиента выполнено подключение:

```
[bahis@client.bahis.net ~]$ ssh -p 2022 bahis@server.bahis.net
The authenticity of host '[server.bahis.net]:2022 ([192.168.1.1]:2022)' can't be established.
ED25519 key fingerprint is SHA256:WWURQh7DVL51N3DFrhBD0VlBovhiQ9Ut00GQzCkE/Xw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[server.bahis.net]:2022' (ED25519) to the list of known hosts.
bahis@server.bahis.net's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Feb 11 14:41:23 2026
[bahis@server.bahis.net ~]$
```

Рисунок 6: Подключение по SSH к серверу через порт 2022 с клиента

- С клиента выполнено подключение:
- `ssh -p 2022 @server.bahis.net`

```
[bahis@client.bahis.net ~]$ ssh -p 2022 bahis@server.bahis.net
The authenticity of host '[server.bahis.net]:2022 ([192.168.1.1]:2022)' can't be established.
ED25519 key fingerprint is SHA256:WWURQh7DVL51N3DFrhBD0VlBovhiQ9Ut00GQzCkE/Xw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[server.bahis.net]:2022' (ED25519) to the list of known hosts.
bahis@server.bahis.net's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Feb 11 14:41:23 2026
[bahis@server.bahis.net ~]$
```

Рисунок 6: Подключение по SSH к серверу через порт 2022 с клиента

- С клиента выполнено подключение:
- `ssh -p 2022 @server.bahis.net`
- Подтверждена успешная авторизация

```
[bahis@client.bahis.net ~]$ ssh -p 2022 bahis@server.bahis.net
The authenticity of host '[server.bahis.net]:2022 ([192.168.1.1]:2022)' can't be established.
ED25519 key fingerprint is SHA256:WWURQh7DVL51N3DFrhBD0VLBovh1Q9Ut00GQzCkE/Xw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[server.bahis.net]:2022' (ED25519) to the list of known hosts.
bahis@server.bahis.net's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Feb 11 14:41:23 2026
[bahis@server.bahis.net ~]$
```

Рисунок 6: Подключение по SSH к серверу через порт 2022 с клиента

- С клиента выполнено подключение:
- `ssh -p 2022 @server.bahis.net`
- Подтверждена успешная авторизация
- Проверена работа перенаправления

```
[bahis@client.bahis.net ~]$ ssh -p 2022 bahis@server.bahis.net
The authenticity of host '[server.bahis.net]:2022 ([192.168.1.1]:2022)' can't be established.
ED25519 key fingerprint is SHA256:WWURQh7DVL51N3DFrhBD0VlBovhiQ9Ut00GQzCkE/Xw.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[server.bahis.net]:2022' (ED25519) to the list of known hosts.
bahis@server.bahis.net's password:
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Wed Feb 11 14:41:23 2026
[bahis@server.bahis.net ~]$
```

Рисунок 6: Подключение по SSH к серверу через порт 2022 с клиента

- Проверен параметр `net.ipv4.ip_forward`

```
[root@server.bahis.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 0
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
net.ipv6.conf.lo.forwarding = 0
net.ipv6.conf.lo.mc_forwarding = 0
[root@server.bahis.net services]# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
[root@server.bahis.net services]# sysctl -p /etc/sysctl.d/90-forward.conf
net.ipv4.ip_forward = 1
[root@server.bahis.net services]# firewall-cmd --zone=public --add-masquerade --permanent
```

- Проверен параметр `net.ipv4.ip_forward`
- Значение — 0 (маршрутизация отключена)

```
[root@server.bahis.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 0
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
net.ipv6.conf.lo.forwarding = 0
net.ipv6.conf.lo.mc_forwarding = 0
[root@server.bahis.net services]# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
[root@server.bahis.net services]# sysctl -p /etc/sysctl.d/90-forward.conf
net.ipv4.ip_forward = 1
[root@server.bahis.net services]# firewall-cmd --zone=public --add-masquerade --permanent
```

- Создан файл /etc/sysctl.d/90-forward.conf

```
[root@server.bahis.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 0
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
```

- Создан файл /etc/sysctl.d/90-forward.conf
- Установлено net.ipv4.ip_forward = 1

```
[root@server.bahis.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 1
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
```

- Создан файл `/etc/sysctl.d/90-forward.conf`
- Установлено `net.ipv4.ip_forward = 1`
- Применена конфигурация `sysctl -p`

```
[root@server.bahis.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 1
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
```

- Создан файл `/etc/sysctl.d/90-forward.conf`
- Установлено `net.ipv4.ip_forward = 1`
- Применена конфигурация `sysctl -p`
- Включён `masquerade` в зоне `public`

```
[root@server.bahis.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 1
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
```

- Создан файл /etc/sysctl.d/90-forward.conf
- Установлено net.ipv4.ip_forward = 1
- Применена конфигурация sysctl -p
- Включён masquerade в зоне public
- Выполнен firewall-cmd --reload

```
[root@server.bahis.net services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 1
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
```


- С клиента выполнены ping 8.8.8.8 и ping google.com

```
[bahis@client.bahis.net ~]$ ping 8.8.8.8.  
ping: 8.8.8.8.: Name or service not known  
[bahis@client.bahis.net ~]$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=254 time=1.40 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=254 time=1.19 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=254 time=0.760 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=254 time=1.36 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=254 time=0.892 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=254 time=0.966 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=254 time=0.969 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=254 time=1.21 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=254 time=1.03 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=254 time=1.49 ms  
64 bytes from 8.8.8.8: icmp_seq=11 ttl=254 time=0.983 ms  
64 bytes from 8.8.8.8: icmp_seq=12 ttl=254 time=1.28 ms  
64 bytes from 8.8.8.8: icmp_seq=13 ttl=254 time=1.40 ms  
64 bytes from 8.8.8.8: icmp_seq=14 ttl=254 time=1.05 ms  
64 bytes from 8.8.8.8: icmp_seq=15 ttl=254 time=0.843 ms  
64 bytes from 8.8.8.8: icmp_seq=16 ttl=254 time=0.755 ms  
64 bytes from 8.8.8.8: icmp_seq=17 ttl=254 time=0.586 ms  
64 bytes from 8.8.8.8: icmp_seq=18 ttl=254 time=1.12 ms  
64 bytes from 8.8.8.8: icmp_seq=19 ttl=254 time=1.20 ms  
64 bytes from 8.8.8.8: icmp_seq=20 ttl=254 time=0.933 ms  
64 bytes from 8.8.8.8: icmp_seq=21 ttl=254 time=0.692 ms  
64 bytes from 8.8.8.8: icmp_seq=22 ttl=254 time=0.991 ms  
64 bytes from 8.8.8.8: icmp_seq=23 ttl=254 time=0.907 ms  
64 bytes from 8.8.8.8: icmp_seq=24 ttl=254 time=0.700 ms  
64 bytes from 8.8.8.8: icmp_seq=25 ttl=254 time=0.770 ms
```

- С клиента выполнены ping 8.8.8.8 и ping google.com
- Потерь пакетов нет

```
[bahis@client.bahis.net ~]$ ping 8.8.8.8.  
ping: 8.8.8.8.: Name or service not known  
[bahis@client.bahis.net ~]$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=254 time=1.40 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=254 time=1.19 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=254 time=0.760 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=254 time=1.36 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=254 time=0.892 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=254 time=0.966 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=254 time=0.969 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=254 time=1.21 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=254 time=1.03 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=254 time=1.49 ms  
64 bytes from 8.8.8.8: icmp_seq=11 ttl=254 time=0.983 ms  
64 bytes from 8.8.8.8: icmp_seq=12 ttl=254 time=1.28 ms  
64 bytes from 8.8.8.8: icmp_seq=13 ttl=254 time=1.40 ms  
64 bytes from 8.8.8.8: icmp_seq=14 ttl=254 time=1.05 ms  
64 bytes from 8.8.8.8: icmp_seq=15 ttl=254 time=0.843 ms  
64 bytes from 8.8.8.8: icmp_seq=16 ttl=254 time=0.755 ms  
64 bytes from 8.8.8.8: icmp_seq=17 ttl=254 time=0.586 ms  
64 bytes from 8.8.8.8: icmp_seq=18 ttl=254 time=1.12 ms  
64 bytes from 8.8.8.8: icmp_seq=19 ttl=254 time=1.20 ms  
64 bytes from 8.8.8.8: icmp_seq=20 ttl=254 time=0.933 ms  
64 bytes from 8.8.8.8: icmp_seq=21 ttl=254 time=0.692 ms  
64 bytes from 8.8.8.8: icmp_seq=22 ttl=254 time=0.991 ms  
64 bytes from 8.8.8.8: icmp_seq=23 ttl=254 time=0.907 ms  
64 bytes from 8.8.8.8: icmp_seq=24 ttl=254 time=0.700 ms  
64 bytes from 8.8.8.8: icmp_seq=25 ttl=254 time=0.770 ms
```

- С клиента выполнены ping 8.8.8.8 и ping google.com
- Потерь пакетов нет
- Подтверждён выход в Интернет через сервер

```
[bahis@client.bahis.net ~]$ ping 8.8.8.8.  
ping: 8.8.8.8.: Name or service not known  
[bahis@client.bahis.net ~]$ ping 8.8.8.8  
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.  
64 bytes from 8.8.8.8: icmp_seq=1 ttl=254 time=1.40 ms  
64 bytes from 8.8.8.8: icmp_seq=2 ttl=254 time=1.19 ms  
64 bytes from 8.8.8.8: icmp_seq=3 ttl=254 time=0.760 ms  
64 bytes from 8.8.8.8: icmp_seq=4 ttl=254 time=1.36 ms  
64 bytes from 8.8.8.8: icmp_seq=5 ttl=254 time=0.892 ms  
64 bytes from 8.8.8.8: icmp_seq=6 ttl=254 time=0.966 ms  
64 bytes from 8.8.8.8: icmp_seq=7 ttl=254 time=0.969 ms  
64 bytes from 8.8.8.8: icmp_seq=8 ttl=254 time=1.21 ms  
64 bytes from 8.8.8.8: icmp_seq=9 ttl=254 time=1.03 ms  
64 bytes from 8.8.8.8: icmp_seq=10 ttl=254 time=1.49 ms  
64 bytes from 8.8.8.8: icmp_seq=11 ttl=254 time=0.983 ms  
64 bytes from 8.8.8.8: icmp_seq=12 ttl=254 time=1.28 ms  
64 bytes from 8.8.8.8: icmp_seq=13 ttl=254 time=1.40 ms  
64 bytes from 8.8.8.8: icmp_seq=14 ttl=254 time=1.05 ms  
64 bytes from 8.8.8.8: icmp_seq=15 ttl=254 time=0.843 ms  
64 bytes from 8.8.8.8: icmp_seq=16 ttl=254 time=0.755 ms  
64 bytes from 8.8.8.8: icmp_seq=17 ttl=254 time=0.586 ms  
64 bytes from 8.8.8.8: icmp_seq=18 ttl=254 time=1.12 ms  
64 bytes from 8.8.8.8: icmp_seq=19 ttl=254 time=1.20 ms  
64 bytes from 8.8.8.8: icmp_seq=20 ttl=254 time=0.933 ms  
64 bytes from 8.8.8.8: icmp_seq=21 ttl=254 time=0.692 ms  
64 bytes from 8.8.8.8: icmp_seq=22 ttl=254 time=0.991 ms  
64 bytes from 8.8.8.8: icmp_seq=23 ttl=254 time=0.907 ms  
64 bytes from 8.8.8.8: icmp_seq=24 ttl=254 time=0.700 ms  
64 bytes from 8.8.8.8: icmp_seq=25 ttl=254 time=0.770 ms
```

2.3 Внесение изменений в настройки внутреннего

- Созданы каталоги:

```
[bahis@client.bahis.net server]$ cd /vagrant/provision/server
[bahis@client.bahis.net server]$ mkdir -p /vagrant/provision/server/firewall/etc/firewalld/services
[bahis@client.bahis.net server]$ mkdir -p /vagrant/provision/server/firewall/etc/sysctl.d
[bahis@client.bahis.net server]$ cp -r /etc/firewalld/services/ssh-custom.xml /vagrant/provision/server/firewall/etc/firewalld/services/
cp: cannot stat '/etc/firewalld/services/ssh-custom.xml': Permission denied
[bahis@client.bahis.net server]$ cp -r /etc/sysctl.d/90-forward.conf /vagrant/provision/server/firewall/etc/sysctl.d/
cp: cannot stat '/etc/sysctl.d/90-forward.conf': No such file or directory
[bahis@client.bahis.net server]$ cd /vagrant/provision/server
[bahis@client.bahis.net server]$ touch firewall.sh
[bahis@client.bahis.net server]$ chmod +x firewall.sh
```

Рисунок 10: Создание структуры каталогов и копирование конфигурационных файлов FirewallD

2.3 Внесение изменений в настройки внутреннего

- Созданы каталоги:
- `firewall/etc/firewalld/services`

```
[bahis@client.bahis.net server]$ cd /vagrant/provision/server
[bahis@client.bahis.net server]$ mkdir -p /vagrant/provision/server/firewall/etc/firewalld/services
[bahis@client.bahis.net server]$ mkdir -p /vagrant/provision/server/firewall/etc/sysctl.d
[bahis@client.bahis.net server]$ cp -r /etc/firewalld/services/ssh-custom.xml /vagrant/provision/server/firewall/etc/firewalld/services/
cp: cannot stat '/etc/firewalld/services/ssh-custom.xml': Permission denied
[bahis@client.bahis.net server]$ cp -r /etc/sysctl.d/90-forward.conf /vagrant/provision/server/firewall/etc/sysctl.d/
cp: cannot stat '/etc/sysctl.d/90-forward.conf': No such file or directory
[bahis@client.bahis.net server]$ cd /vagrant/provision/server
[bahis@client.bahis.net server]$ touch firewall.sh
[bahis@client.bahis.net server]$ chmod +x firewall.sh
```

Рисунок 10: Создание структуры каталогов и копирование конфигурационных файлов FirewallD

2.3 Внесение изменений в настройки внутреннего

- Созданы каталоги:
- `firewall/etc/firewalld/services`
- `firewall/etc/sysctl.d`

```
[bahis@client.bahis.net server]$ cd /vagrant/provision/server
[bahis@client.bahis.net server]$ mkdir -p /vagrant/provision/server/firewall/etc/firewalld/services
[bahis@client.bahis.net server]$ mkdir -p /vagrant/provision/server/firewall/etc/sysctl.d
[bahis@client.bahis.net server]$ cp -r /etc/firewalld/services/ssh-custom.xml /vagrant/provision/server/firewall/etc/firewalld/services/
cp: cannot stat '/etc/firewalld/services/ssh-custom.xml': Permission denied
[bahis@client.bahis.net server]$ cp -r /etc/sysctl.d/90-forward.conf /vagrant/provision/server/firewall/etc/sysctl.d/
cp: cannot stat '/etc/sysctl.d/90-forward.conf': No such file or directory
[bahis@client.bahis.net server]$ cd /vagrant/provision/server
[bahis@client.bahis.net server]$ touch firewall.sh
[bahis@client.bahis.net server]$ chmod +x firewall.sh
```

Рисунок 10: Создание структуры каталогов и копирование конфигурационных файлов FirewallD

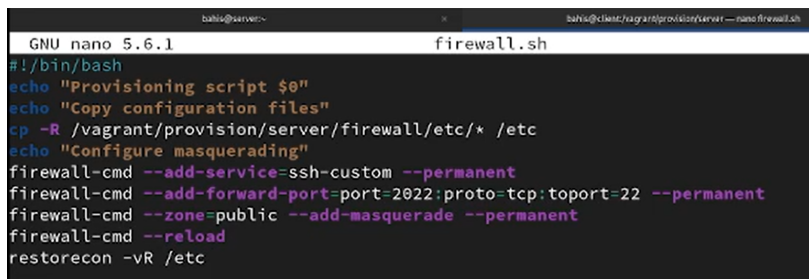
2.3 Внесение изменений в настройки внутреннего

- Созданы каталоги:
- `firewall/etc/firewalld/services`
- `firewall/etc/sysctl.d`
- Скопированы `ssh-custom.xml` и `90-forward.conf`

```
[bahis@client.bahis.net server]$ cd /vagrant/provision/server
[bahis@client.bahis.net server]$ mkdir -p /vagrant/provision/server/firewall/etc/firewalld/services
[bahis@client.bahis.net server]$ mkdir -p /vagrant/provision/server/firewall/etc/sysctl.d
[bahis@client.bahis.net server]$ cp -r /etc/firewalld/services/ssh-custom.xml /vagrant/provision/server/firewall/etc/firewalld/services/
cp: cannot stat '/etc/firewalld/services/ssh-custom.xml': Permission denied
[bahis@client.bahis.net server]$ cp -r /etc/sysctl.d/90-forward.conf /vagrant/provision/server/firewall/etc/sysctl.d/
cp: cannot stat '/etc/sysctl.d/90-forward.conf': No such file or directory
[bahis@client.bahis.net server]$ cd /vagrant/provision/server
[bahis@client.bahis.net server]$ touch firewall.sh
[bahis@client.bahis.net server]$ chmod +x firewall.sh
```

Рисунок 10: Создание структуры каталогов и копирование конфигурационных файлов FirewallD

- Создан скрипт `firewall.sh`

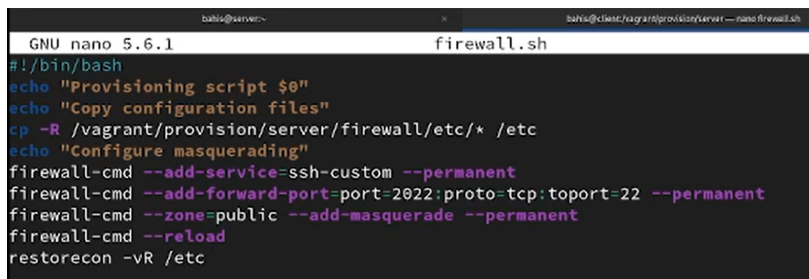


The screenshot shows a terminal window with two tabs. The active tab is titled `bahts@client:vagrant/provision/server — nano firewall.sh`. The terminal displays the content of the `firewall.sh` script, which is being edited in `nano 5.6.1`. The script starts with a shebang `#!/bin/bash` and contains several `echo` statements and `firewall-cmd` commands. The commands configure the firewall to allow SSH, forward port 22, and reload the configuration. The script ends with `restorecon -vR /etc`.

```
GNU nano 5.6.1 firewall.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 11: Содержимое provisioning-скрипта `firewall.sh`

- Создан скрипт `firewall.sh`
- Реализовано:

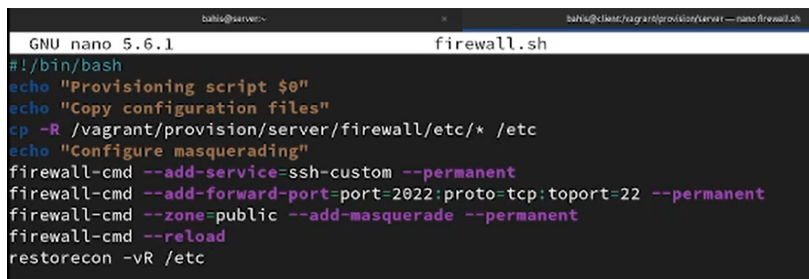


The screenshot shows a terminal window with two tabs. The active tab is titled `bahts@client:vagrant/provision/server — nano firewall.sh`. The terminal displays the content of the `firewall.sh` script, which is being edited in `nano` version 5.6.1. The script starts with a shebang `#!/bin/bash` and contains several `echo` statements for logging, followed by `cp` and `firewall-cmd` commands to configure the firewall.

```
GNU nano 5.6.1 firewall.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 11: Содержимое provisioning-скрипта `firewall.sh`

- Создан скрипт `firewall.sh`
- Реализовано:
- копирование конфигураций в `/etc`

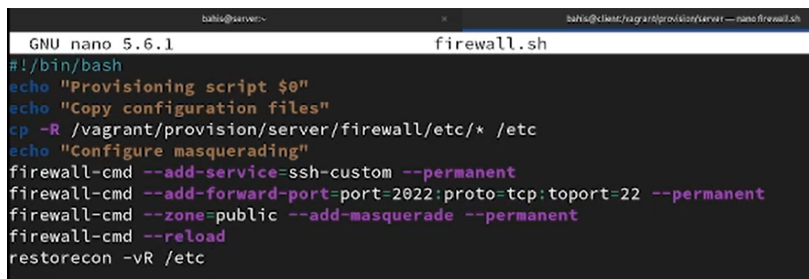


The screenshot shows a terminal window with the title bar "bahis@server:~" and "GNU nano 5.6.1 firewall.sh". The script content is as follows:

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 11: Содержимое provisioning-скрипта `firewall.sh`

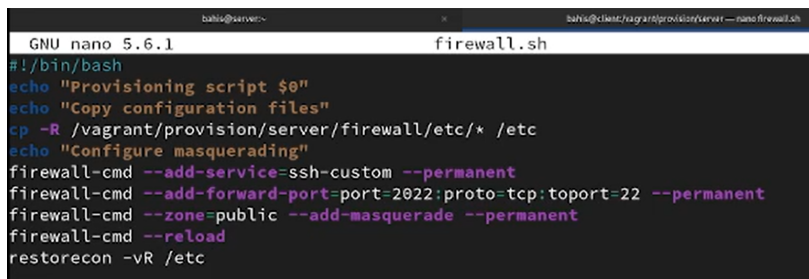
- Создан скрипт `firewall.sh`
- Реализовано:
- копирование конфигураций в `/etc`
- добавление службы `ssh-custom`



```
GNU nano 5.6.1 firewall.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 11: Содержимое provisioning-скрипта `firewall.sh`

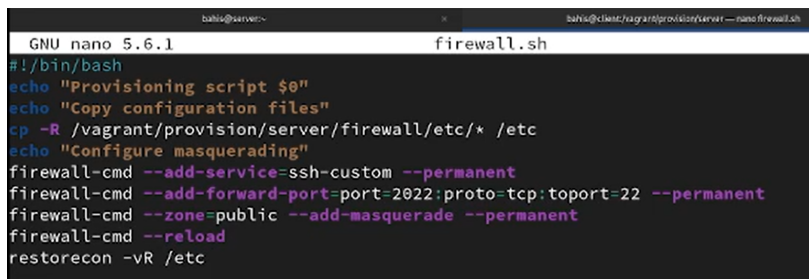
- Создан скрипт `firewall.sh`
- Реализовано:
- копирование конфигураций в `/etc`
- добавление службы `ssh-custom`
- перенаправление `2022` → `22`



```
GNU nano 5.6.1 firewall.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 11: Содержимое provisioning-скрипта `firewall.sh`

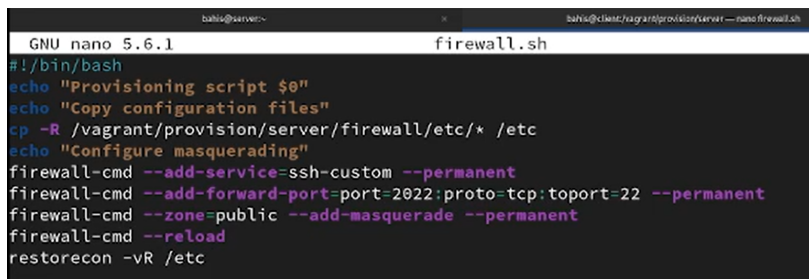
- Создан скрипт `firewall.sh`
- Реализовано:
- копирование конфигураций в `/etc`
- добавление службы `ssh-custom`
- перенаправление `2022` → `22`
- включение `masquerade`



```
GNU nano 5.6.1 firewall.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 11: Содержимое provisioning-скрипта `firewall.sh`

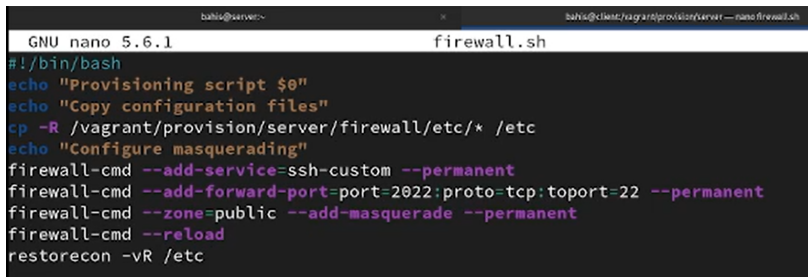
- Создан скрипт `firewall.sh`
- Реализовано:
- копирование конфигураций в `/etc`
- добавление службы `ssh-custom`
- перенаправление `2022` → `22`
- включение `masquerade`
- перезагрузка `firewalld`



```
GNU nano 5.6.1 firewall.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 11: Содержимое provisioning-скрипта `firewall.sh`

- Создан скрипт `firewall.sh`
- Реализовано:
- копирование конфигураций в `/etc`
- добавление службы `ssh-custom`
- перенаправление `2022` → `22`
- включение `masquerade`
- перезагрузка `firewalld`
- восстановление SELinux-контекстов



```
GNU nano 5.6.1 firewall.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рисунок 11: Содержимое provisioning-скрипта `firewall.sh`

- В Vagrantfile добавлен provision-блок

```
server.vm.provision "server firewall",  
type: "shell",  
preserve_order: true,  
path: "provision/server/firewall.sh"
```

Рисунок 12: Добавление provisioning-скрипта firewall.sh в Vagrantfile

- В Vagrantfile добавлен provision-блок
- Тип shell

```
server.vm.provision "server firewall",  
type: "shell",  
preserve_order: true,  
path: "provision/server/firewall.sh"
```

Рисунок 12: Добавление provisioning-скрипта firewall.sh в Vagrantfile

- В Vagrantfile добавлен provision-блок
- Тип shell
- Путь provision/server/firewall.sh

```
server.vm.provision "server firewall",  
type: "shell",  
preserve_order: true,  
path: "provision/server/firewall.sh"
```

Рисунок 12: Добавление provisioning-скрипта firewall.sh в Vagrantfile

- В Vagrantfile добавлен provision-блок
- Тип shell
- Путь provision/server/firewall.sh
- Указан preserve_order: true

```
server.vm.provision "server firewall",  
type: "shell",  
preserve_order: true,  
path: "provision/server/firewall.sh"
```

Рисунок 12: Добавление provisioning-скрипта firewall.sh в Vagrantfile

Раздел 3

3. Выводы

3.1 Выводы

- Создана пользовательская служба `ssh-custom` (порт 2022)

3.1 Выводы

- Создана пользовательская служба `ssh-custom` (порт 2022)
- Реализовано перенаправление `2022 → 22`

3.1 Выводы

- Создана пользовательская служба `ssh-custom` (порт 2022)
- Реализовано перенаправление `2022 → 22`
- Подтверждён SSH-доступ через нестандартный порт

3.1 Выводы

- Создана пользовательская служба `ssh-custom` (порт 2022)
- Реализовано перенаправление `2022 → 22`
- Подтверждён SSH-доступ через нестандартный порт
- Включена маршрутизация IPv4 (`ip_forward = 1`)

3.1 Выводы

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